

Virtual-Reality and Behavioural Studies: A Useful Tool For Testing Passenger Wayfinding In Railway Station

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ABSTRACT

This study presents a virtual-reality-based assessment of three alternative wayfinding systems in a medium-size French regional railway station. Using a high-fidelity VR replica of the station, 20 participants completed three navigation scenarios under three signage configurations: suspended signage, a continuous colored ground line, and a control configuration. Eye-tracking, dwell-time metrics, and subjective data were collected. Results show that both suspended signage and the colored line significantly improved navigation efficiency and structured gaze patterns. Findings seem to support deploying a hybrid system combining overhead and ground guidance in comparable stations.

INDEX TERMS: Virtual reality, eye tracking, wayfinding, behavioural studies, public transportation.

1 INTRODUCTION

Passenger wayfinding in regional railway stations depends heavily on the clarity, consistency, and spatial redundancy of signage systems. The station studied in this project is a medium-size French railway station, found in small cities, which features a remote platform positioned away from the main circulation path. In this station, this particular layout frequently leads to navigational difficulties.

To address these issues, the station operator commissioned a virtual-reality-based evaluation of several alternative wayfinding concepts. Virtual reality provides a controlled and cost-effective means of testing signage before physical deployment, and is widely used in navigation and human-factors research for its ecological validity (Riecke et al., 2010).

The addition of integrated eye-tracking makes VR especially suitable for studying moment-to-moment perceptual strategies (Duchowski, 2017). Because the environment closely replicates real-world conditions, stakeholders can explore design alternatives at full scale and assess their feasibility before committing to physical implementation. The goals were to:

- Evaluate the user experience in VR through the dimensions of presence, engagement and immersion;
- Compare the effectiveness of three candidate wayfinding systems;
- Identify visual strategies used during navigation to the remote platform;
- Produce operational recommendations for deployment in comparable stations.

2 METHOD

2.1 Participants

Twenty adults (10 women, 10 men; aged 21–64, $M = 37.2$) participated. Their familiarity with the station ranged from none to frequent use, and train-use frequency varied from rare to weekly.

2.2 VR Environment and Setup

A high-fidelity VR replica of the station was created, including the main hall, tunnels, stairways, escalators, elevators, and four platforms. Ambient audio and virtual passengers enhanced realism. Participants wore an HTC Vive Pro Eye headset measuring locomotion, gaze fixations (≥ 100 ms), and head-orientation angles.



Figure 1: Participant with VR helmet

2.3 Signage Conditions

Three systems were evaluated:

- Suspended signage: repeated overhead signs in the tunnel, at hall transitions, and along the remote platform.
- Colored ground line: a continuous, high-contrast path linking all entrances to the platform, supplemented by metric markers.
- Control condition (baseline): existing signage in the station. In this configuration, only the elevator layout was modified in order to see the effect on participant behaviour.



Figure 2: Colored ground line in the main hall

2.4 Procedure

Participants completed three realistic navigation scenarios involving tasks such as meeting a friend, reacting to platform changes, or guiding another traveler. Scenario and signage order were counterbalanced. Total session time was 90 minutes.

2.5 Measures

Collected data are dwell times across station zones; gaze-angle distributions; fixation counts and durations for signage elements; user experience questionnaire in VR (IVEQ) and qualitative interviews.

3 RESULTS

3.1 VR Presence

Participants found the VR station realistic and reported navigating as they would in a real environment. Presence was significantly correlated with engagement (Spearman $r = .537$, $p = .018$), indicating that greater immersion supported more attentive behavior. These findings are consistent with the ecological validity demonstrated in prior VR navigation research (Riecke et al., 2010).

3.2 Quantitative Analysis

3.2.1 Dwell Time

Participants navigated more efficiently with the enhanced systems. In the main hall, dwell time decreased from 26 seconds in the baseline to 20 seconds with the colored line, indicating quicker orientation. In the tunnel, traversal was fastest with suspended signage, improving times by roughly 12% compared to the control condition.

3.2.2 Gaze-Angle Distribution

The colored line increased downward gaze, i.e. the lower visual field in front of the feet (about 40%), showing reliance on ground cues. This highlights the importance of a visible/continuous signage, which appears to delineate a path perceived as having low congestion. Suspended signage concentrated gaze at eye level (slightly above 50%), demonstrating its role as an initial reference. The control condition produced a less structured distribution.

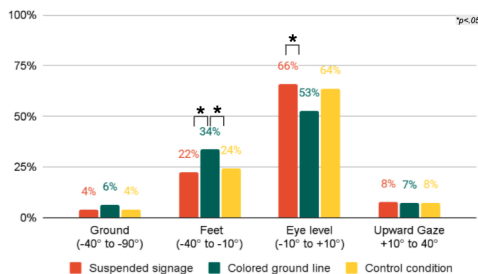


Figure 3: Example of gaze distribution in the tunnel

3.2.3 Fixation Metrics

The colored line was fixated in 100% of passages and revisited multiple times, confirming its strong perceptual salience. These results should be interpreted within the context of an experimental setting devoid of visibility-obstruction (i.e., no crowding and no dirt). Suspended signs in the tunnel drew frequent fixations (around 15 per traversal). By contrast, baseline directional panels were fixated in only about one-third of passages.

3.3 Qualitative Analysis

Nearly all participants reported that they first looked for suspended signs as their primary source of direction. They then used the ground markings to confirm the route, frequently

describing the colored line as “reassuring” and “unambiguous.” So, 95% of participants suggest that the best solution is a mix of the colored line and suspended signage.

4 DISCUSSION

The results highlight the strength of the suspended signs to facilitate initial orientation and the interest of the colored line to provide continuous reassurance. These findings must be interpreted cautiously, as they were obtained under experimental conditions without visibility-masking factors, such as crowding or surface contamination.

This study exemplifies how VR and behavioral sciences can jointly inform design decisions, offering stakeholders concrete, evidence-based support for implementing or refining wayfinding solutions. This combination of experimental rigor and immersion aligns with the growing literature demonstrating VR’s value for studying complex environments (Chen et al., 2018).

5 CONCLUSION

The next step would be to potentially increase the participant sample in order to confirm these observations, and subsequently to conduct a real-world comparison, either using a simple scenario or within one of the spaces where significant differences were observed.

Finally, it would be valuable to approach real station conditions more closely by increasing occupancy levels and assessing whether the colored line remains salient for travelers.

ACKNOWLEDGEMENTS

Authors would like to thank The Region Centre-Val de Loire, the French State and the BpiFrance, for their financial support of the Virtual Lab project, through the France 2030 program.

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